sample_4-Evaluation report

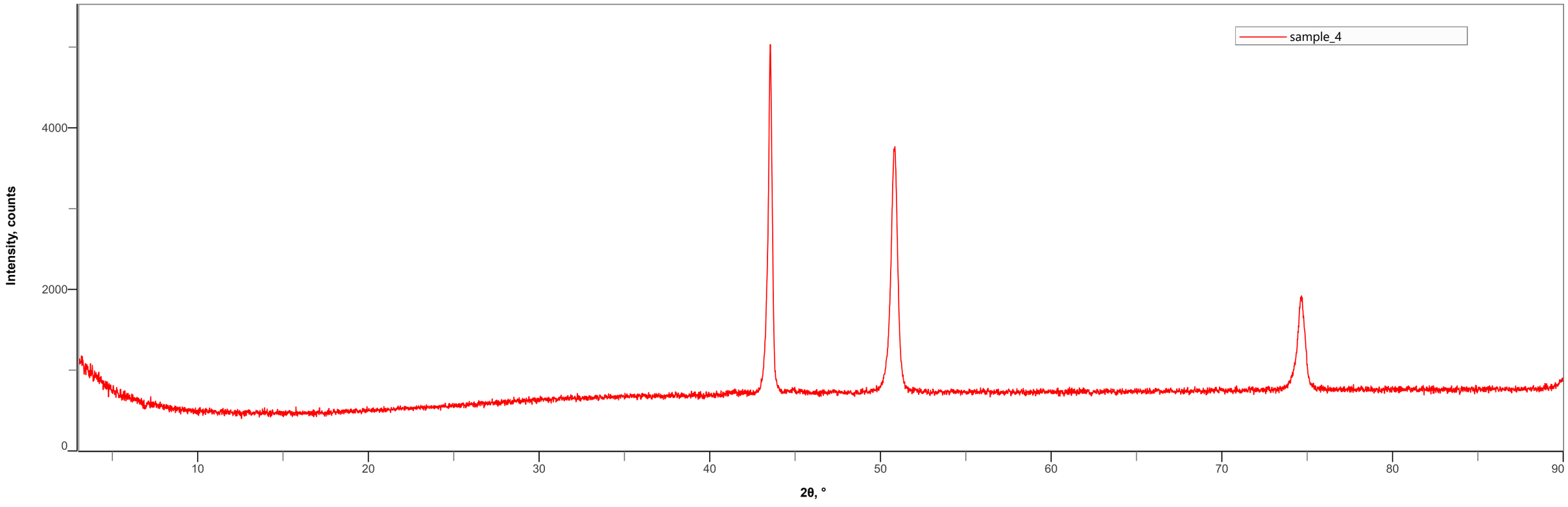
General information

Analysis date	2025-11-27 13:26:18	Measurement start time	2025-11-23 13:12:37
Analyst	Administrator	Operator	Administrator
Sample name	sample_4	Comment	
Measured data name	C:\Users\DELL\Desktop\251\Vasanth\sample_4.rasx	Memo	

Measurement Conditions

X-Ray generator	40 kV, 15 mA	Scan mode	1D(scan)
Incident primary	No unit	Scan speed/Duration time	10.00 °/min
Goniometer	MiniFlex 300/600	Step width	0.01 °
Attachment	Standard sample stage	Scan axis	θ/2θ
Filter	None	Scan range	3 ~ 90 °
Selection slit	None	DS	1.25deg
Diffracted beam mono	None	IHS	10 mm
Detector	HyPix-400 MF	SS	Open
Optics attribute	None	RS	Open

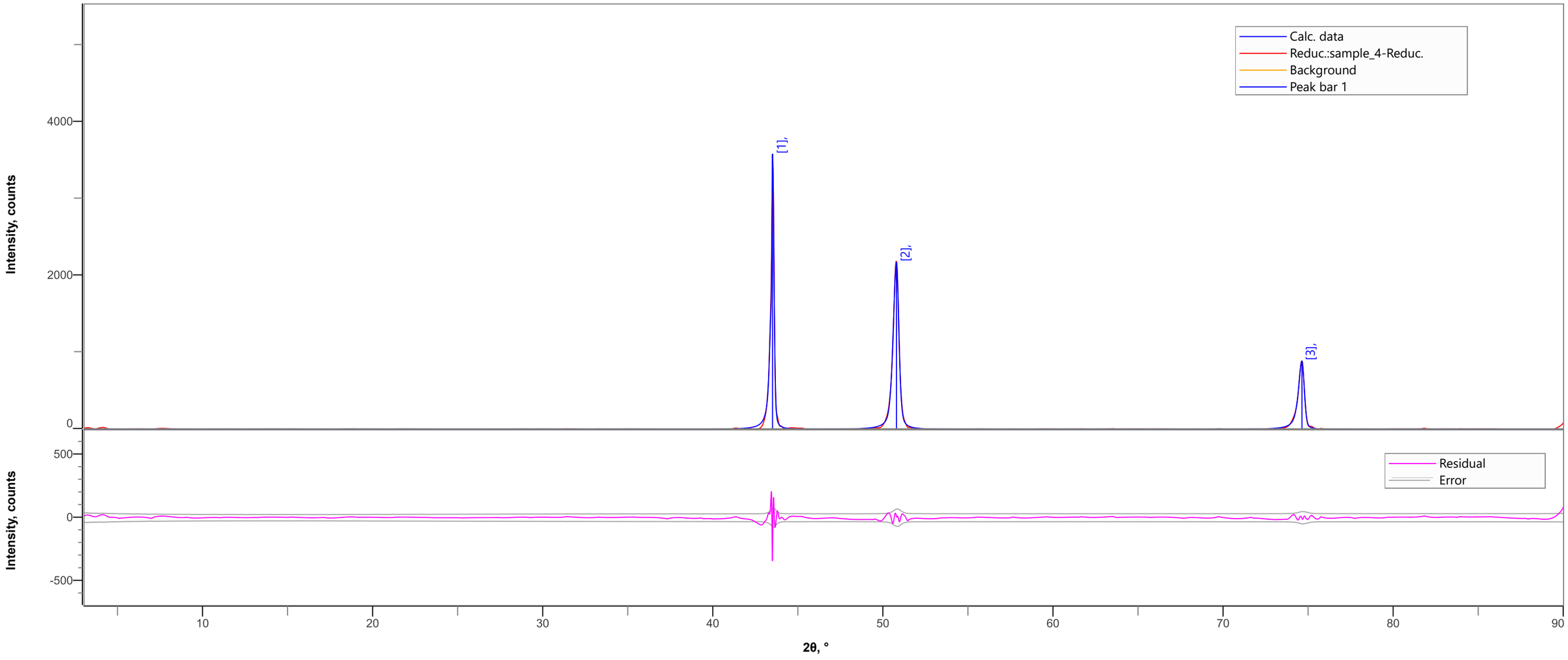
Measured profile view



Peak evaluation conditions

Peak search method	Second derivative method	σ cut	35.00		
Profile fitting	Run completed	Peak shape	Split pseudo-Voigt	Fitting condition	Auto(Refine background)

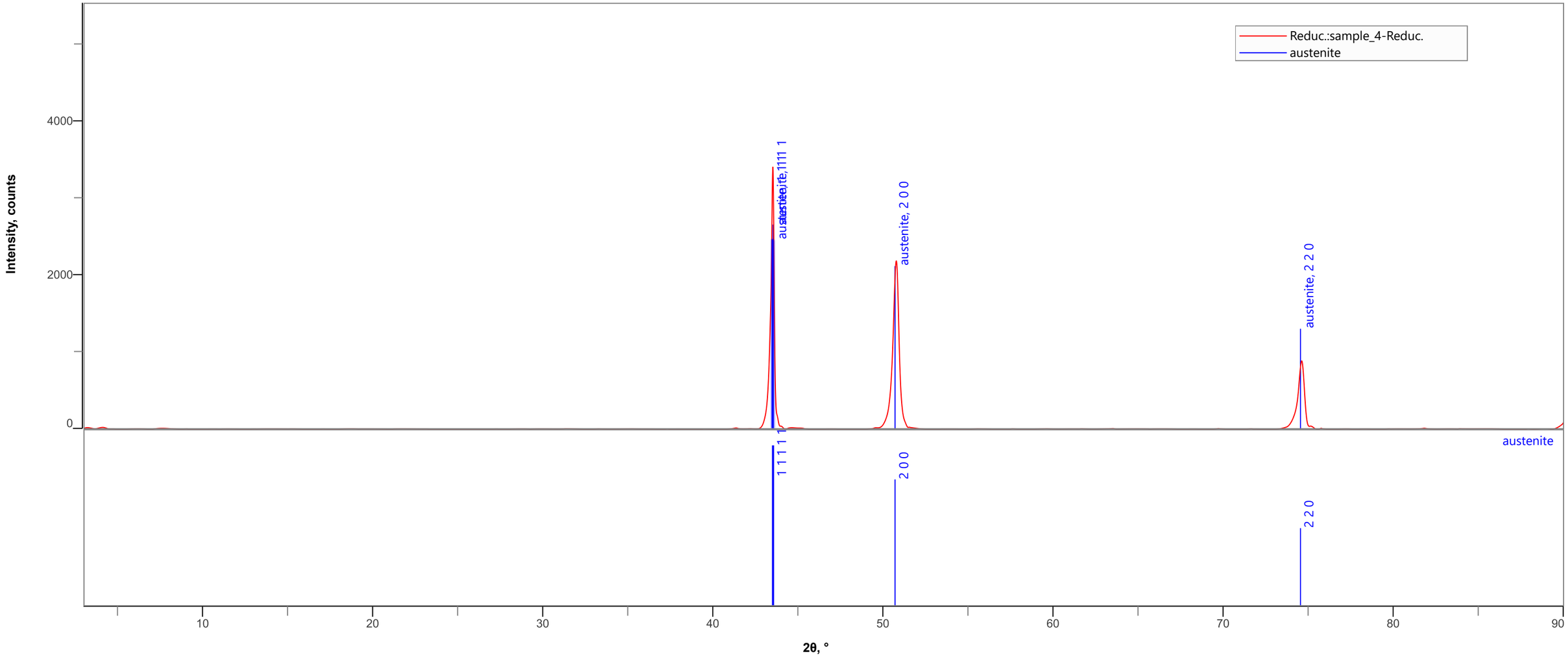
Peak Profile View



Qualitative Analysis Results

Phase name	Chemical formula	FOM	Phase reg. detail	Space Group	DB Card Number
austenite	Cr - Ni - Fe - C	0.364	S/M:PDF-5+ 2026	225 : Fm-3m	00-031-0619

Phase Data View



Peak list

No.	2θ, °	d, Å	Height, counts	FWHM, °	Int. I., counts°	Int. W., °	Asymmetry	Decay (ηL/mL)	Decay(ηH/mH)	Size, Å	Phase Name	Chemical Formula	Card No	Norm. I.	Profile Type	Distribution R.S.D.	Strain, %	Degree of Orientation	Ring Factor	β Cluster	Custom Label
1	43.48795(14)	2.079298(6)	3578(81)	0.1704(3)	967.4(11)	0.270(6)	0.7247(19)	1.539(2)	0.364(5)	524.2(8)	austenite: 1 1 1	Cr - Ni - Fe - C	00-031-0619	89.37	Split Pseudo-Voigt	-	-	-	-	-	
2	50.7686(3)	1.796870(9)	2171(67)	0.3900(6)	1082.5(12)	0.498(16)	1.312(3)	0.550(4)	0.476(5)	235.5(3)	austenite: 2 0 0	Cr - Ni - Fe - C	00-031-0619	100.00	Split Pseudo-Voigt	-	-	-	-	-	
3	74.5949(5)	1.271215(8)	889(49)	0.4013(11)	479.4(10)	0.54(3)	1.479(8)	0.788(6)	0.400(11)	259.9(7)	austenite: 2 2 0	Cr - Ni - Fe - C	00-031-0619	44.29	Split Pseudo-Voigt	-	-	-	-	-	

Lattice parameters

Phase name	a, Å	b, Å	c, Å	α, °	β, °	γ, °
austenite	3.59912	3.59912	3.59912	90.000	90.000	90.000

d-I List

austenite

No.	2θ, °	d, Å	h k l	Norm. I.
1	43.51754	2.07795	1 1 1	100.00
2	50.68737	1.79956	2 0 0	80.00
3	74.50806	1.27248	2 2 0	50.00

Simple Inverse Pole Figure

austenite

